

Functional Annotation of PP_3205 (UniProt Q88HZ7) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

PP_3205 (Q88HZ7) encodes FdeI, a soluble cytoplasmic enzyme of the fumarylacetoacetate-hydrolase (FAH) superfamily (hydratase/decarboxylase family; COG0179) that acts in the bacterial catabolic pathway for the plant flavonoid naringenin (and related flavanones/flavones). By orthology and KEGG pathway assignment, its product participates in the lower, "meta-cleavage" portion of the *fde* (flavonoid-degradation) pathway — the steps that process the ring-fission products generated after dioxygenolytic cleavage of the flavonoid A-ring — funnelling the carbon skeleton toward oxaloacetate and the TCA cycle. It carries out its reaction in the **cytoplasm**. It is expressed from a **flavonoid-inducible operon** controlled by the LysR/NodD-like regulator FdeR.

Its precise catalyzed reaction and substrate specificity have **not been determined experimentally for PP_3205 itself**; the functional assignment is an inference from (i) orthology to the experimentally characterized *Herbaspirillum seropedicae* SmR1 pathway, (ii) conserved genomic clustering, (iii) KEGG orthology (K26185 / FdeI), and (iv) the diagnostic FAH-fold catalytic architecture in its sequence.

2. Gene/Protein Identity Verification

Attribute	Value
Locus tag	PP_3205 (ordered locus name; unambiguous in KT2440)
UniProt	Q88HZ7 (330 aa)
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Genome position	3,638,005–3,638,997 (+ strand); 993 nt CDS
Protein family	Fumarylacetoacetate hydrolase (FAH) superfamily; "hydratase/decarboxylase family" (UniProt SIMILARITY)
Domains	FAA_hydro_N (IPR041072 / Pfam PF18288) + FAA_hydrolase C-terminal (IPR011234, fold IPR036663 / Pfam PF01557)
Orthology	eggNOG COG0179 ; KEGG ortholog K26185 (fdeI)
KEGG pathway	ppu00946 "Degradation of flavonoids"
GenBank protein	AAN68813.1

Verification outcome: The locus tag, organism, domain architecture (two-domain FAH fold) and family assignment are mutually consistent. The generic UniProt name ("Fumarylacetoacetate hydrolase family protein") is correct but uninformative; the *specific* functional identity comes from KEGG orthology (FdeI, naringenin degradation) and the experimentally defined *H. seropedicae* pathway. No conflicting "same-symbol, different-gene" literature issue arises because PP_3205 is an ordered-locus identifier, not an ambiguous common symbol.

3. Primary Function

3.1 What the enzyme is

FdeI belongs to the **fumarylacetoacetate-hydrolase (FAH) superfamily**, whose members are divalent-metal-dependent (typically $\text{Ca}^{2+}/\text{Mg}^{2+}$) enzymes that catalyze **C–C bond hydrolysis, hydration, or decarboxylation of oxo-carboxylate intermediates**. Canonical examples include fumarylacetoacetase (tyrosine catabolism), fumarylpyruvate hydrolase (gentisate/naphthalene catabolism, e.g., NagK/SgpK;

P 11133965

P 23266498

and 2-keto-4-pentenoate hydratase (aromatic meta-cleavage pathways). The UniProt "hydratase/decarboxylase family" designation reflects this chemistry.

3.2 Pathway context and the specific step

In the experimentally dissected *Herbaspirillum seropedicae* SmR1 pathway (Marin et al. 2016,

P 27059806

1. **FdeE** (a monooxygenase) performs the initial mono-oxygenation of naringenin to 5,7,8-trihydroxy-2-(4-hydroxyphenyl)-2,3-dihydro-4H-chromen-4-one. 2. **FdeC** (a ring-cleaving dioxygenase) then **meta-cleaves the A-ring**. 3. Downstream enzymes (including the FAH-family FdeI) process the ring-opened products; the pathway ultimately **generates oxaloacetate, which enters the tricarboxylic acid cycle**.

FdeI (PP_3205) is one of these downstream, ring-fission-processing enzymes. KEGG links K26185/FdeI to reactions R13078 (a decarboxylation: 5-hydroxy-5-(4-hydroxyphenyl)-3-oxopentanoate → 4-hydroxy-4-(4-hydroxyphenyl)butan-2-one + CO_2) and R13079 (a hydration/dehydration). **These two reactions are flagged "unclear reaction" in KEGG and should be treated as provisional:** they capture the *type* of chemistry expected of a FAH-family enzyme (decarboxylation + hydration of an oxo-carboxylate) but the exact substrate/product for FdeI have not been confirmed enzymologically.

3.3 Substrate specificity

No direct kinetic/substrate data exist for PP_3205. Consistent with the whole operon being induced by **apigenin**, **chrysin**, **luteolin** and **naringenin** (Wassem et al. 2017,

P 27878922

), the pathway — and therefore FdeI — most plausibly acts on ring-fission intermediates derived from **several flavanones/flavones**, not naringenin alone.

3.4 Genomic clustering in KT2440 (evidence for pathway membership)

PP_3205 lies within a contiguous, co-oriented cluster whose members carry catabolic annotations matching a flavonoid meta-cleavage pathway:

Locus	Annotation	KEGG ortholog
PP_3201	BNR-repeat domain protein	—
PP_3202 / PP_3203	transporter / MFS transporter	K07003 / —
PP_3204	conserved protein	K26182 (FdeH)
PP_3205	FAH-family protein	K26185 (FdeI)
PP_3206	conserved protein	K26183 (FdeJ)
PP_3207	putative cyclase	—
PP_3208	Pdr/VanB-family oxidoreductase	—
PP_3212	Rieske 2Fe-2S dioxygenase	—

The presence of a Rieske dioxygenase, an oxidoreductase, a cyclase, transporters and three Fde orthologs in one locus is the hallmark of an aromatic-compound catabolic island, corroborating the KEGG assignment. The K26185 ortholog group is broadly conserved across *Pseudomonas* (e.g., PVLB_09785, PSEBR_a2142, T1E_5649).

4. Subcellular Localization

The gene product functions in the **cytoplasm**: - No N-terminal signal peptide (sequence begins MKLATFDDGS...) and no transmembrane segment: whole-protein GRAVY = -0.023 (hydrophilic) and the maximum Kyte–Doolittle hydropathy over a 19-residue window is only 1.04, below the ~ 1.6 threshold indicative of a membrane helix. - FAH-superfamily enzymes are characteristically soluble cytoplasmic proteins, and the naringenin ring-fission intermediates they process are generated intracellularly. - UniProt provides no subcellular-location annotation, consistent with a default cytoplasmic assignment.

5. Structure / Mechanistic Inference

- **Fold:** Two-domain FAH architecture — an N-terminal FAA_hydro_N cap (PF18288) plus the C-terminal catalytic FAA_hydrolase domain (PF01557, IPR036663 superfamily fold).
- **Catalytic center:** The sequence contains the conserved His-rich C-terminal motif HLPLHVHWNGEWFHGPHGG (His217/His221/His223/His231/His233 with adjacent Glu/Asp). In FAH-superfamily structures, an equivalent His/Glu-rich pocket coordinates a divalent metal ion and activates a water nucleophile for C–C bond cleavage/hydration/decarboxylation. The presence of an intact motif argues the protein is a **catalytically competent** family member rather than a degenerate pseudoenzyme.
- An AlphaFold model is available (AlphaFoldDB: Q88HZ7) and would be expected to reproduce the canonical FAH β -sandwich fold.

6. Biological Process / Physiological Role

FdeI contributes to **flavonoid catabolism**, enabling the bacterium to use plant-derived flavonoids as carbon and energy sources. Because the operon is specifically induced by flavonoids via the NodD-like LysR regulator FdeR (Wassem et al. 2017,

P 27878922)

— a regulator otherwise associated with rhizobial *nod*-gene control — the pathway is best understood as an adaptation to **plant-associated / rhizosphere niches**, where flavonoids are

abundant plant exudates. The end product (oxaloacetate) is assimilated through central metabolism (Marin et al. 2016,).

P 27059806

7. Supported vs. Refuted Hypotheses

Supported (by convergent evidence): - H1 — PP_3205 is a FAH-superfamily hydrolase/decarboxylase. (UniProt/Pfam/InterPro/COG0179 + intact catalytic motif.) - H2 — PP_3205 = FdeI and functions in the naringenin/flavonoid meta-cleavage pathway. (KEGG K26185/ppu00946 + conserved gene cluster + orthology to characterized *H. seropedicae* pathway,).

P 27059806

- H3 — The enzyme is a soluble cytoplasmic protein. (No signal peptide/TM; hydropathy analysis.) - H4 — The pathway (and thus fdeI) is flavonoid-inducible. (

P 27878922

Refuted / not supported: - The protein is **not** a membrane transporter or secreted protein (no TM/signal). The neighbouring transporters (PP_3202/3203, PP_3210/3211) perform substrate import, not FdeI. - No evidence that PP_3205 acts in tyrosine catabolism (classic fumarylacetoacetase role) in KT2440; that reaction is served by the separate homogentisate pathway. The FAH fold here is repurposed for flavonoid ring-fission chemistry.

8. Limitations and Future Directions

- **No direct experimental characterization of PP_3205 exists.** Function is inferred from orthology (chiefly to *H. seropedicae*), KEGG, and sequence. The exact FdeI substrate/product and kinetics are unknown; the KEGG-listed reactions R13078/R13079 are provisional ("unclear reaction").
- The precise ordering of the lower-pathway steps (which of FdeH/FdeI/FdeJ acts first, and the true chemical structures of the intermediates) is not firmly established.
- **Recommended experiments:** heterologous expression and in-vitro assay of PP_3205 with candidate ring-fission substrates; Δ PP_3205 growth phenotype on naringenin/apigenin/

chrysin/luteolin; metabolite (LC-MS/MS) profiling of the mutant; and an experimental structure or docking of intermediates into the AlphaFold model to identify the metal ligands and active-site residues.

9. Key References

- Marin AM, de la Torre J, Oliveira ARM, et al. *Genetic and functional characterization of a novel meta-pathway for degradation of naringenin in Herbaspirillum seropedicae SmR1*. Environ Microbiol. 2016;18(12):4653–4661.

P 27059806

- Wassem R, Marin AM, Daddaoua A, et al. *A NodD-like protein activates transcription of genes involved with naringenin degradation in a flavonoid-dependent manner in Herbaspirillum seropedicae*. Environ Microbiol. 2017;19(3):1030–1040.

P 27878922

- Zhou N-Y, Fuenmayor SL, Williams PA. *nag genes of Ralstonia sp. strain U2 encoding enzymes for gentisate catabolism*. J Bacteriol. 2001;183(2):700–708.

P 11133965

(fumarylpyruvate hydrolase NagK — FAH-family precedent).

- Izmalkova TY, et al. *The organization of naphthalene degradation genes in Pseudomonas putida strain AK5*. Res Microbiol. 2013.

P 23266498

(fumarylpyruvate hydrolase SgpK — FAH-family precedent).

- Database records: UniProt Q88HZ7; KEGG ppu:PP_3205, K26185, pathway ppu00946; Pfam PF01557/PF18288; InterPro IPR041072/IPR011234/IPR036663; eggNOG COG0179.